

Skills Summary

Piecewise and Absolute-Value Models — Boundaries and Endpoints

Use this reference while completing your worksheet or when studying.

Skill 1 — Which rule owns the boundary?

The rule with the closed inequality owns the boundary.

- $\leq$ and $\geq$ **include** their endpoint; $<$ and $>$ **exclude** it.
- Test the input against the *conditions* first, pick the one rule that is true, and only then substitute.
- Either rule may be the closed one — top or bottom. There is no convention to memorise, so read the sign every time.

Example: For $P(x) = \begin{cases} 2x + 4, & x < 4 \\ x^2 - 10, & x \geq 4 \end{cases}$ find $P(4)$.

Step 1. The input is the boundary, so decide ownership before substituting: $4 < 4$ is false and $4 \geq 4$ is true, so the bottom rule owns it.

Step 2. Substitute into the bottom rule: $4^2 - 10$.

$P(4) = 6$. The top rule's value there would have been 12, which is what P approaches from the left but never takes.

Try it: For $Q(x) = \begin{cases} 6 - x, & x \leq 1 \\ x + 6, & x > 1 \end{cases}$ find $Q(1)$.

check: 5

Skill 2 — Overlaps, gaps, and matching values

Every input must be owned by exactly one rule.

- Both conditions closed at the boundary $\rightarrow$ **overlap**: two outputs for one input, so it is not a function. Open one side.
- Both strict $\rightarrow$ **gap**: the boundary has no output and is outside the domain. Close one side.
- **No jump** is a separate question: set the two formulas equal *at* the boundary and solve for the unknown constant. Matching values does not change which rule owns the point.

Example: Find k so that $R(x) = \begin{cases} 2x + k, & x < 5 \\ x^2 - 9, & x \geq 5 \end{cases}$ has no jump at its boundary.

Step 1. A jump is a mismatch of values at one input, so set the two rules equal at the boundary and solve for the constant — ownership is already settled by the signs.

Step 2. Set $2(5) + k = 5^2 - 9$ and solve for k .

$k = 6$, and the bottom rule still owns $x = 5$.

Try it: Find k so that $S(x) = \begin{cases} 3x + k, & x < 2 \\ x^2 + 4, & x \geq 2 \end{cases}$ has no jump.

check: $\lim_{x \rightarrow 2^-} S(x) = 6 + k$

Skill 3 — Endpoints of an absolute-value range

Solve the equation to find the endpoints; read the sign to include them.

- $|A| = c$ splits into $A = c$ and $A = -c$. Those two solutions are the endpoints of the region $|A| \leq c$.
- $|A| \leq c$ keeps both endpoints (closed); $|A| < c$ drops both (open). The endpoints are the same numbers either way.
- In a tolerance context, the closed form accepts a part that hits the limit exactly — reporting an open range rejects parts that meet the specification.

Example: Give the endpoints of the region $|3x - 6| \leq 9$ and say whether they belong to it.

Step 1. The boundary of an absolute-value region is where the expression equals the limit, so solve the equation $|3x - 6| = 9$ rather than guessing from a graph.

Step 2. Split into $3x - 6 = 9$ and $3x - 6 = -9$, then solve each.

$x = -1$ or $x = 5$. The sign is $\leq$, so both endpoints belong and the region is $-1 \leq x \leq 5$.

Try it: Give the endpoints of the region $|2x + 1| \leq 7$.

check: $\lim_{x \rightarrow -1} (2x + 1) = -1$